Cambridge (CIE) A Level Computer Science

Your notes

Data Types & Records

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Primitive data types

What are data types?

- A data type is a classification of data into groups according to the **kind of data they represent**
- Computers use different data types to represent different types of data in a program
- The basic data types include:
  - **Integer**: used to represent **whole numbers**, either positive or negative
    - Examples: 10, -5, 0
  - **Real**: used to represent **numbers with a fractional part**, either positive or negative
    - Examples: 3.14, -2.5, 0.0
  - **Char**: used to represent a **single character** such as a letter, digit or symbol
    - Examples: 'a', 'B', '5', '$'
  - **String**: used to represent a **sequence of alphanumerical characters**
    - Examples: "Hello World", "1234", "@#$%"
  - **Boolean**: used to represent **true or false** values
    - Examples: True, False
  - **Date**: used to store a **calendar date**
    - Example: DD/MM/YY
- It is important to choose the **correct data type for a given situation** to ensure accuracy and efficiency in the program
- In pseudocode and some other programming languages, **data types must be declared** before they can be used
  - The data types are declared with each **data item** to be used
  - Each data item is given a **unique** name called an **identifier**
  - DECLARE <identifier> : <data type>

Data Type	Example Value	Pseudocode	Python	Java	VB.NET
Integer	10	DECLARE Age : INTEGER Age ← 10	age = 10	int age = 10;	Dim age As Integer = 10

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Real	3.14	DECLARE Temp : REAL Temp ← 3.14	temp = 3.14	double temp = 3.14;	Dim temp As Double = 3.14
Char	'A'	DECLARE Grade : CHAR Grade ← 'A'	grade = 'A'	char grade = 'A';	Dim grade As Char = "A"c
String	"Hello"	DECLARE Name : STRING Name ← "Hello"	name = "Hello"	String name = "Hello";	Dim name As String = "Hello"
Boolean	TRUE	DECLARE LoggedIn : BOOLEAN LoggedIn ← TRUE	logged_in = True	boolean loggedIn = true;	Dim loggedIn As Boolean = True
Date	24/04/2025	DECLARE DOB : DATE DOB ← "24/04/2025"	dob = "24/04/2025" (string or datetime)	LocalDate dob = LocalDate.of(2025, 4, 24);	Dim dob As Date = #24/04/2025#

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Purpose of records

What is a record?

- A record is a **composite data structure** used to store a **collection of related data items**
- Each item **can be** of a **different data type**
- It allows a programmer to group related values under **one identifier**
- Records help create a **more structured and readable** approach to storing and managing data
- A record contains a **fixed number of fields**, each with its own name and type
- In pseudocode, a record can by declared as:

```
TYPE <Typename>
   DECLARE <identifier> : <data type>
   DECLARE <identifier> : <data type>
   DECLARE <identifier> : <data type>
   ..
   ..
ENDTYPE
```

- Suppose we want to store information about a car
- We can define a **record type** like this:

```
TYPE Car
   DECLARE Make : STRING
   DECLARE Model : STRING
   DECLARE Colour : STRING
   DECLARE Price : REAL
   DECLARE DateOfRegistration : DATE
ENDTYPE
```

- We can then **create a variable** of this type:

```
DECLARE MyCar : Car
```

- And **assign values** to its fields:

```
MyCar.Make ← "Toyota"
MyCar.Model ← "Yaris"
MyCar.Colour ← "Red"
MyCar.Price ← 14995.99
MyCar.DateOfRegistration ← "12/03/2022"
```

- To **read from the record** we can:

```
OUTPUT "Make: ", MyCar.Make
OUTPUT "Model: ", MyCar.Model
OUTPUT "Colour: ", MyCar.Colour
OUTPUT "Price: £", MyCar.Price
OUTPUT "Date of Registration: ", MyCar.DateOfRegistration
```


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Examiner Tips and Tricks

- **Records** in programming are a type of **data structure** used to group related data in your code
- These are **not the same** as records in a database, which refer to a **complete set of fields** on a single entity in a table (**row**)

Key features of records

Feature	Explanation
Can store **different data types**	Unlike arrays, records are not limited to a single type
Fields are **named**	Each item in the record has a meaningful identifier
Offers **structured storage**	Easier to manage complex data about real-world entities

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